WASTEWISE SCAN · CONCEPT OPTIONS

WasteWise Scan concept options

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DOCUMENT CONTROL

Document ID	WSC-OPT-001	Version	0.1
Status	Draft	Date	2026-10-24
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wastewise-scan		
Source	docs/02a-concept-options.md @ 8c61176		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First concept options (kit 1.8 upgrade): option A is the current design (with fixes F1 to F3); option B, a tunable-filter spectrometer, added; both modelled at D3 (kit 1.9); scores within 10 %, so both go to Amish without a pick

WasteWise Scan concept options

Option A, the current eight-band LED scanner, scores 186 of 235 against 173 for option B, a tunable-filter spectrometer: A is ahead by 7.0 %, inside the 10 % band, so **both go to Amish without a pick** (STANDARDS section 21). A is cheaper, faster and tougher; B reads a continuous spectrum across the 1,650 to 1,765 nm polymer bands that A only reaches at their short edge, which is where PE against PP is at risk (PR-001). The repo keeps the current design as option A, as the standard requires for a design that predates kit 1.8; nothing is redesigned without Amish's go. Method: STANDARDS section 21. Selection record: WSC-DDR-005.

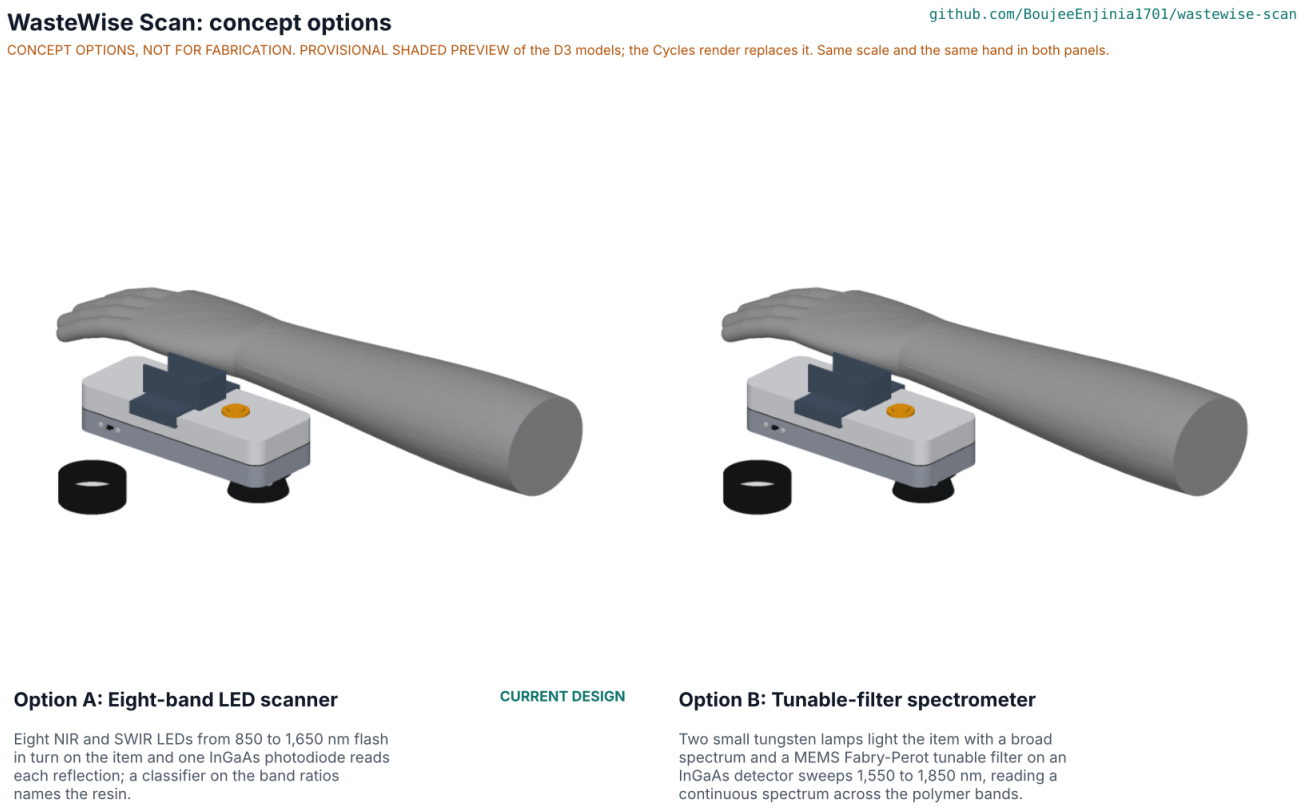


Figure 1. The two options, each a D3 model from `cad/src/model.py` *(* `components("A")`, `components("B")` *), drawn at one scale with a hand for scale. The picture is rendered with* `python .kit/options.py` *in Blender Cycles; until then the file is a labelled shaded preview.*

The options

Option A: eight-band LED scanner (current design)

Eight NIR and SWIR LEDs from 850 to 1,650 nm flash in turn on the item and one InGaAs photodiode reads each reflection; a classifier on the band ratios names the resin. This is the constructable design of WSC-DDR-004 with fixes F1 to F3 (baffle through the glass, four-cycle scan, larger cell strap; WSC-DEC-001). Each band is read between two dark readings, four times over; clear items in sun are backed with the black calibration cap. First-order numbers from WSC-CAL-001: 0.64 s per scan, signal-to-noise ratio of 640 or more in shade, 21.9 h per charge, 239 g (259 g with the cap), USD 167.50 in parts. Key risks: PE against PP (the 1,650 nm band is clipped by the detector

cutoff; paper band study 93 % for that pair, 89.5 % for all five resins at 1.5 % item scatter), display legibility in direct sun, and SWIR LED prices. From a co-design partner it needs a labelled set of local items (200 or more) and the local price grades.

Option B: tunable-filter spectrometer

Two small tungsten lamps light the item with a broad spectrum, and a MEMS Fabry-Perot tunable filter on an InGaAs detector, in one TO-5 class can, sweeps 1,550 to 1,850 nm and reads a continuous spectrum (Hamamatsu C13272 series range, cited in the sources). It shares A's body, shroud, split window, display board, cell and cap; only the optical head changes: a taller head block holds the sensor in its centre bore and the two lamps on the body axis, tilted about 28° in aluminium sleeves (fix F4) so their light meets on the item, and a driver board above it carries the filter's tuning supply, the amplifier and the converter. First-order numbers (WSC-STM-001 S30 to S36, estimates): about 0.74 s per scan for a 100-point sweep (before any repeat cycles for sunlight like option A's), signal-to-noise ratio well above 200 per 20 nm step, 19.3 h per charge, about 242 g (262 g with the cap), and at least USD 375 in parts. Key risks: the sensor price (INR 26,640 excluding taxes for one sensor on IndiaMART, about USD 300 at an assumed INR 88 per USD), lamp glass heat next to a PETG head block, filament life in drops, the filter's settling time and its voltage-to-wavelength calibration, and the loss of the 1,200 nm and 1,400 nm bands that help with PET and PVC. From a co-design partner it needs the same labelled item set, and a supplier who can sell single sensors in India.

First-order numbers

Table 1. Options compared.

Quantity	Option A	Option B
Estimated parts cost (USD)	167.50 (priced BOM, WSC-CAL-001 K2)	At least 375 (A's BOM less USD 92 of LEDs and photodiode, plus about USD 300 for the sensor; lamps and filter driver to assess)
Mass (g)	239; 259 with the cap (WSC-CAL-001 A3)	About 242; 262 with the cap (estimate: head block 5 g heavier, optics 2 g lighter, sleeves about 1 g)
Output or throughput	0.64 s per scan, 8 bands, four cycles (E2)	About 0.74 s per scan, about 100 points from 1,550 to 1,850 nm (S33)
Energy or effort	0.40 W average, 21.9 h per charge (F2)	About 0.45 W average, 19.3 h per charge (S35)
Parts count (D3 model, `cad/fidelity.json`)	86 parts: 48 made or bought, 38 fasteners	78 parts: 40 made or bought, 38 fasteners
Mass in the D3 model (kit densities, summed from `components()`)	218.8 g	220.5 g
STEM check	No failed row after fixes F1 to F3; 11 of 23 rows plausible	No failed row after fix F4; 3 of 7 rows plausible

The model masses use the kit's material densities, which have no PETG, TPU or PTFE (HDPE stands in), and leave out wire and solder; the calculation note's 239 g is the design figure.

Scoring

Table 2. Weighted scoring (weights: High UR 3, Medium 2, Low 1; scores 1 to 5).

Criterion	Source	Weight	A	B	Reason
Right resin	UR-001	3	3	4	B covers the PE and PP first-overtone bands fully; A reaches their short edge (93 % on paper for PE against PP); B loses the 1,200 nm band
Quick	UR-002	3	4	4	0.64 s against about 0.74 s, both well under 1.5 s
Honest unknown	UR-003	3	4	4	Same confidence rule and black threshold
Works in the sun	UR-004	3	4	4	Both need the backing step for clear items; sunlight still outweighs either source in a 20 nm slice
Lasts a shift	UR-005	3	5	4	21.9 h against 19.3 h
One hand, light	UR-006	3	3	3	Both about 9 to 12 g over 250 g with the cap
No reading needed	UR-007	3	3	3	Same display and sun hood
Our price grade	UR-008	3	5	5	Same firmware feature
A record of scans	UR-009	2	5	4	B's spectrum record is larger (about 100 points), fewer scans per partition, still far over 10,000
Survives handling	UR-010	3	4	3	Solid-state LEDs against tungsten filaments and a MEMS filter in a 1,200 g corner drop
Easy to keep right	UR-011	2	4	3	B adds lamp colour drift with cell voltage and a wavelength calibration of the filter
Affordable to a cooperative	UR-012	3	3	1	USD 167.50 (12 % over the target) against at least USD 375 (2.5 times)
Works with the phone app	UR-013	2	4	4	Same shared record
Safe invisible light	UR-014	3	4	5	A needs a hardware on-time limit on invisible LEDs; B's lamps glow visibly and are far below the limits
Safe battery	UR-015	3	4	4	Same cell, cradle and strap; B draws a higher peak current
Built locally	UR-016	2	4	3	B needs a tuning supply of tens of volts and a calibrated filter driver
Our data, our consent	UR-017	2	5	5	Same record and consent terms
Maintenance		1	4	3	Lamps wear out and need replacing; LEDs outlast the device
Weighted total (of 235)			**186**	**173**	A ahead by 7.0 %: inside 10 %, so no pick

Cost against the value-engineering target, buildability and safety are scored through UR-012, UR-016, UR-014 and UR-015.

Recommendation

No pick: the two options are within 10 %, so both go to Amish as **"Proposed, awaiting Amish"** (WSC-DEC-001, open decision 2; WSC-DDR-005). The repo keeps option A as its design until Amish decides, as the standard requires, and `concept_chosen` stays unset in `project.yaml`.

What each choice means:

- **Choose A** (the current design): no change to the repo; R1's PE and PP risk is settled by the first labelled item set at TRL 4. The runner-up's best idea to graft onto A is B's reach into the 1,700 to 1,750 nm band, as a ninth LED band with an extended InGaAs photodiode, which is the open decision 1 already in WSC-DEC-001.
- **Choose B**: the optical head, its board, the BOM, the build plan and the CAL change; the value-engineering target would be missed by at least USD 225, and the sensor would need a confirmed single-unit supplier. The best idea to keep from A is its 1,200 nm band, as one LED beside the lamps, for PET and PVC.

Sources

- Hamamatsu C13272 series MEMS-FPI spectrum sensor, 1,550 to 1,850 nm, tunable filter and InGaAs PIN photodiode in one package: Photonics Online, https://www.photonicsonline.com/doc/ultra-small-mems-fpi-spectrum-sensor-c-datasheet-0001; Photonics.com, https://www.photonics.com/Product.aspx?PRID=61532 (TO-5 can). Both checked 2026-10-24.
- IndiaMART listing, Hamamatsu C13272-02 MEMS-FPI spectrum sensor, INR 26,640 per piece excluding taxes, range given there as 1,550 to 1,850 nm. https://m.indiamart.com/proddetail/mems-fpi-spectrum-sensor-23949601312.html. Checked 2026-10-24. The exchange rate of about INR 88 per USD is assumed, to confirm when bought; the exact type number and its range are to confirm with Hamamatsu before any purchase (to assess).
- WasteWise Scan calculation note WSC-CAL-001 v0.5 and STEM check WSC-STM-001 v0.1 for every other number.